

VANESSA SKLYAROVA

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Computer Vision & AI researcher with 6+ years of experience working on diverse research & engineering projects. At the moment, a Ph.D. student at ETH Zurich, ([CVG group](#)) and Max Planck Institute for Intelligent Systems ([Perceiving Systems](#)) with a focus on digital humans. Five first-author papers at top venues including **ICCV 2023 (Oral, top 1.8%)** and **3DV 2026 (Best Paper Runner-Up)**.

PROFESSIONAL EXPERIENCE

Meta Zurich, ([Reality Labs](#))

Research Scientist Intern

06/2026 - Present

- Research on video diffusion models.

ETH Zurich, ([CVG group](#))

Ph.D. Researcher

07/2025 - Present

- Research on dynamic head avatars from multi-view images.

Max Planck Institute for Intelligent Systems, ([Perceiving Systems](#))

Ph.D. Researcher

06/2023 - Present

- 3D reconstruction, rendering, and simulation of head avatars with strand-based hair.
- Developed a text-guided generative model and single-view strand reconstruction method.
- Research on hair rendering with Gaussian Splatting.
- Published 4 first-author papers (CVPR 2024, ICCV 2025, 3DV 2026, ECCV 2026) + 7 collaborative works.
- Supervised 5 students and 4 group projects on computer vision and virtual reality.

Samsung AI Center

Research Engineer

08/2021 - 06/2023

- Worked on mesh reconstruction methods for learning human hair geometry;
- Co-authored a paper accepted to ECCV 2022 on one-shot mesh reconstruction of human avatars;
- Co-authored a patent on reconstruction and rendering of human head avatars ([#12169900](#));
- Worked on multi-view hair reconstruction methods with strand-based representations;
- Co-authored a paper accepted to ICCV 2023 on strand-based hair reconstruction for human avatars.

Samsung AI Center

Assistant Engineer

06/2021 - 08/2021

- Developed a Reptile-based meta-learning framework for NeRF models;
- Integrated the meta-learning pipeline into the top-performing model in distributed mode.

Huawei

Assistant Engineer

06/2020 - 10/2020

- Tested state-of-the-art low-bit quantization methods on Transformer and CNN/MLP models;
- Conducted experiments on automatic quantization using reinforcement learning.

Gazprom Neft

Assistant Engineer

02/2019 - 01/2020

- Built RL agents (DQN, DQfD) and contextual bandits for dynamic pricing, achieving 90%+ of optimal rewards;
- Co-authored patent on dynamic pricing ([#2022619566](#));

EDUCATION

ETH Zurich

Ph.D. in Computer Science

Zurich, Switzerland

09/2023 - Present

Topic: *Digital human avatars*, prof. Marc Pollefeys, prof. Michael Black, prof. Justus Thies

Skolkovo Institute of Science and Technology, [Skoltech](#)

Moscow, Russia

M.Sc. in Data Science (GPA 4.84 / 5.00)

Topic: *Reinforcement learning for Deep brain simulation* ([thesis](#)), prof. Dmitry Dyllov

Moscow Institute of Physics and Technology, [MIPT](#)

Moscow, Russia

M.Sc. in Applied Mathematics and Physics (GPA 4.95 / 5.00)

Moscow Institute of Physics and Technology, [MIPT](#)

Moscow, Russia

B.Sc. in Applied Mathematics and Physics (GPA 4.4 / 5.00)

Topic: *Mathematical model of HIV infection*, Institute of Numerical Mathematics, prof. Gennady Bocharov

RELEVANT PUBLICATIONS & PREPRINTS

Head Avatars with Dynamic Explicit Hair

2026

V. Sklyarova*, H. Chen*, B. Kabadayi, T. Kirschstein, Z. Fan, X. Wang, G. Pons-Moll, M. Nießner, M. Pollefeys, M. Black, J. Thies

ECCV

MOCHI: Registration-free Multi-view Face Prediction ([page](#))

2026

P. Filntisis, G. Retsinas, R. Danecek, [V. Sklyarova](#), P. Maragos, T. Bolkart

CVPR

PhysHead: Simulation-Ready Gaussian Head Avatars ([page](#))

2026

B. Kabadayi, [V. Sklyarova](#), W. Zielonka, J. Thies, G. Pons-Moll

CVPR

GeomHair: Reconstruction of Hair Strands from Colorless 3D Scans ([page](#))

2026

R. Lazuardi, A. Sevastopolsky, E. Zakharov, M. Nießner, [V. Sklyarova](#)

CVPRW

How to Build Digital Humans? From Priors to Photorealistic Avatars ([page](#))

2026

W. Zielonka*, T. Kirschstein*, T. Bolkart*, S. Giebenhain, [V. Sklyarova](#), X. Deng, et al.

Eurographics (STAR)

Vid2Haircut: 3D Strand-Based Hairstyle Reconstruction from Video ([page](#))

2026

F. Ben Ayed, G. Becherini, J. Thies, [V. Sklyarova](#)

Eurographics (Short Papers), CVPRW

NeuralFur: Animal Fur Reconstruction from Multi-view Images ([page](#))

2026

[V. Sklyarova*](#), B. Kabadayi*, A. Yiannakidis, G. Becherini, M. Black, J. Thies

3DV

(Oral presentation, Best Paper Runner Up Award)

Im2Haircut: Single-view Strand-based Hair Reconstruction for Human Avatars ([page](#))

2025

[V. Sklyarova](#), E. Zakharov, M. Prinzler, G. Becherini, M. Black, J. Thies

ICCV

Joker: Conditional 3D Head Synthesis with Extreme Facial Expressions ([page](#))

2025

M. Prinzler, E. Zakharov, [V. Sklyarova](#), B. Kabadayi, J. Thies

3DV

Gaussian Haircut: Human Hair Reconstruction with Strand-Aligned 3D Gaussians ([page](#))

2024

E. Zakharov, [V. Sklyarova](#), M. Black, G. Nam, J. Thies, O. Hilliges

ECCV

HAAR: Text-Conditioned Generative Model of 3D Strand-based Human Hairstyles ([page](#))

2024

[V. Sklyarova](#), E. Zakharov, O. Hilliges, M. Black, J. Thies

CVPR

Neural Haircut: Prior-Guided Strand-Based Hair Reconstruction ([page](#))

2023

[V. Sklyarova](#), J. Chelishev, A. Dogaru, I. Medvedev, V. Lempitsky, E. Zakharov

ICCV

(Oral presentation, top 1.8% of the submissions)

Realistic one-shot mesh-based head avatars ([page](#))

2022

T. Khakhulin, [V. Sklyarova](#), V. Lempitsky, E. Zakharov

ECCV

* denotes equal contribution.

SELECTED AWARDS

Best paper runner-up award <i>NeuralFur: Animal Fur Reconstruction from Multi-view Images</i>	Vancouver, Canada March 2026
Outstanding Reviewer Award	CVPR 2026
Best presentation runner-up award <i>Text-Conditioned Generative Model of 3D Strand-based Human Hairstyles</i>	Erlangen, Germany Dec 2023

SERVICE & COMMUNITY CONTRIBUTIONS

Workshop Organizer. MUSTCV Workshop, ECCV 2026; CV4Metaverse Workshop, CVPR 2025.
Conference Reviewer. CVPR, ICCV, ECCV, SIGGRAPH, SIGGRAPH Asia, 3DV, Eurographics.
Student Supervision. Supervised 5 M.Sc. students, 1 B.Sc. student, and 4 group projects for the Computer Vision and Virtual Reality course at ETH Zurich. Co-authored two student-led papers as senior author (Vid2Haircut, GeomHair).

TEACHING

3D Vision , ETH Zurich, CVG Lab	Spring 2026
Linear Algebra , ETH Zurich	Autumn 2025
Seminar on Human-Performance Capture , ETH Zurich, AIT Lab	2024 (2 semesters)

PATENTS

Reconstruction and Rendering of Human Head Avatars, U.S. Patent ([#12169900](#)). Co-inventor.
Dynamic Pricing System, Russian Patent ([#2022619566](#)). Co-inventor.

SELECTED PRESENTATIONS

Keynote <i>"Modeling Strand-based Hairstyles for Digital Human Avatars"</i>	P3HA workshop CVPR 2026
Oral Presentation <i>"NeuralFur: Animal Fur Reconstruction From Multi-View Images"</i>	Vancouver, Canada 3DV 2026
Invited Talk <i>"Modeling Realistic Hairstyles for Digital Human Avatars"</i>	ETH AI Center Jan 2026
Talk <i>"Text-Conditioned Generative Model of 3D Strand-based Human Hairstyles"</i>	FAU University, Erlangen Dec 2023
Invited Talk <i>"Generating Realistic 3D Hair Geometry for Human Avatars"</i>	CVG Lab, ETH Zurich May 2024
Invited Talk <i>"Generating Realistic 3D Hair Geometry for Human Avatars"</i>	Google Zurich Feb 2024
Oral Presentation <i>"Neural Haircut: Prior-Guided Strand-based Hair Reconstruction"</i>	Paris, France ICCV 2023
Talk <i>"Neural Haircut: Prior-Guided Strand-based Hair Reconstruction"</i>	CV4Metaverse Workshop ICCV 2023
Talk <i>"Neural Haircut: Prior-Guided Strand-based Hair Reconstruction"</i>	Amazon, Tübingen July 2023

LANGUAGES

English (full professional), German (basic), Russian (native)

SKILLS / AREAS

Digital Humans · 3D Modeling and Reconstruction · 3D Scanning · 2D & 3D Computer Vision · 3DMMs · NeRFs · Gaussian Splatting · Neural Rendering · Diffusion Models · Hair and fur reconstruction · Physics-based simulations · Computer Graphics for Vision · Reinforcement Learning · Python · C/C++ · PyTorch · Blender